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ChatGPT Assistance in Indonesian Argumentative Writing: Effects Across Dimensions and Initial Performance

Author(s)

Abstract

Generative artificial intelligence (GenAI) has been widely recognized to significantly assist students in writing high-quality essays. Yet the tool's effectiveness in improving academic writing in languages other than English is still limited. This study addresses this gap by examining the effectiveness of ChatGPT assistance in improving Indonesian argumentative essays. The study further examined the writing dimensions and the group of learners based on their initial writing performance, who significantly benefited from GenAI support. Using a quantitative design, the study collected Indonesian argumentative essays from 52 undergraduate students, grouped into those with and without ChatGPT assistance. Two expert raters assessed the essays using the ESL Composition Profile developed by Jacobs et al. Two sets of data from between-group and within-group comparisons, analyzed using Welch's t-tests and Hedges' g and paired-samples t-tests and Pearson correlation analyses, respectively, revealed that ChatGPT assistance remains significant for writing an Indonesian argumentative essay. Among the writing dimensions assessed, content, organization, and language use showed the greatest improvement with p value from $<.001$ – 0.044 and Hedge's g from 0.614-0.81, whereas vocabulary and mechanics showed limited gains, particularly between group comparisons with p value from 0.334-0.352 and Hedge's g from 0.263-0.265. Further, improvement was negatively correlated with students' initial writing performance, indicating that lower-performing writers gain more benefits from using ChatGPT as a writing assistant than their higher-performing peers. The findings suggest that ChatGPT can serve as an effective pedagogical scaffold to enhance higher-order aspects of argumentative writing, particularly for less proficient writers.

Keywords: GenAI, ChatGPT, Academic Writing, Indonesian Argumentative Essays, Jacob's ESL Composition Profile

INTRODUCTION

Academic writing is critically important for university students as a competence that enables them to engage in academic discussion through constructive and critical thinking and arguments (Petric, 2002). However, writing a piece of academic writing is often reported as challenging due to students' unfamiliarity with writing conventions and limited writing skills, not to mention when writing in an English as a Foreign Language (EFL) context (Badi, 2015; Hartshorn et al., 2023; A. Mustafa et al., 2013; Nghi & Nhan, 2023; Ningrum et al., 2023). Hence, scholars and educators are integrating current technological innovations to support students in academic writing, among which

⁵ GenAI tools, such as ChatGPT, have become a favorable option.

Systematic and scoping reviews have consistently shown that integrating ChatGPT into writing courses yields benefits. Reviewing 70 empirical studies, Lo et al. (2024) found converging evidence that ChatGPT not only facilitates idea generation, essay organization, vocabulary development, and grammatical accuracy but also improves overall writing quality. Similar findings were reported across studies examining students' perspectives, in which students credited ChatGPT for their better drafts and improved grammar and vocabulary (Misbah Sultan et al., 2025; Nguyen et al., 2024). The likely pattern was reported in Indonesia, where university students could expand their vocabulary, generate better ideas, improve sentence

structure, and enhance the overall quality of their essays with ChatGPT's assistance (Werdiningsih et al., 2024). Synthesizing 20 empirical studies, Teng (2024) found that ChatGPT not only supports EFL students through the multiple stages of the writing process, from brainstorming, drafting, revision, and feedback, but also encourages learner autonomy, writing confidence, and self-regulated learning. Meanwhile, Mustafa et al. (2024)'s synthesis of 50 review articles concludes that GenAI's significance lies in its ability to provide personalized learning, immediate formative feedback, and adaptive learning support, thereby enhancing writing instruction.

More empirical research in English as a Second Language (ESL) and EFL contexts increasingly suggests that ChatGPT can enhance students' academic writing performance. Studies across different EFL context, including Vietnamese, Thai, Chinese, Saudi, Pakistani, and Indonesian found that ChatGPT-assisted writing interventions have generally been associated with improvements in students' overall writing performance as well as specific dimensions of essay quality, despite uniformly distributed across all writing components (Khampusaen, 2025; Mahapatra, 2024; Mai, 2026; Mekheimer, 2025; Mujeeb et al., 2026; Puteri et al., 2025; Q. Xiao, 2024). ChatGPT's influence is more significant when interventions last for six to sixteen weeks, as improvements in language-related and structural dimensions, including better argument organization, more appropriate vocabulary, improved sentence accuracy, and more effective writing revision, become more consistent. A similar effect was observed among Korean EFL students who received ChatGPT-assisted feedback. The students were reported to improve their writing quality in content, organization, and language accuracy better than their peers who did not have ChatGPT's assistance but peer feedback (Mun, 2024). Another similar finding was reported in an examination of Saudi Arabian EFL students' writing quality after a three-week ChatGPT intervention in the writing class (Alangari, 2025). The students' improvement was observed in their essays across content, vocabulary, and language use, though modest

improvement remained in organization and mechanics. However, the encouraging findings should not be interpreted to mean that higher essay scores necessarily indicate competence, as students may not improve in writing ability (Herbold et al., 2023) and may not write well without ChatGPT's intervention, leading to a high reliance on GenAI (Tossell et al., 2024).

Further, the literature on ChatGPT's adoption in academic writing reveals another important pattern regarding its influence on learners of varying proficiency levels. Most studies reported that ChatGPT's intervention in the writing process is more significant for students of lower proficiency, as found among EFL learners (Mekheimer, 2025). The significance is most observable in students' improved grammar, vocabulary, content, and organization in their writing compared to peers with higher proficiency. A similar tendency is reported by Puteri et al. (2025), who found that lower-proficiency Indonesian EFL students showed greater improvement after ChatGPT-supported instruction. However, Puteri et al. (2025) further reported that the students' improvement was not significantly greater than that of students receiving traditional writing instruction, indicating that the generally positive evidence for ChatGPT's effectiveness should not be overstated.

Despite increasing interest and growing evidence on ChatGPT-assisted writing, the existing literature poses several important gaps. The first gap is that empirical research has been dominated by perception-based approaches, in which perceptions are collected through questionnaires, interviews, or students' self-reported experiences rather than objective evaluations of students' writing outcomes (Alangari, 2025; Khampusaen, 2025; Mahapatra, 2024; Mai, 2026; Mekheimer, 2025; Mujeeb et al., 2026; Mun, 2024; Puteri et al., 2025; Q. Xiao, 2024). Consequently, the extent to which ChatGPT influences the actual quality of students' written work remains largely underexplored (Albadarin et al., 2024). In addition, most existing empirical studies assess students' writing development after several weeks of ChatGPT-assisted instruction. In contrast, fewer studies examine students' essays written under ChatGPT-assisted

conditions, with no precondition for ChatGPT intervention. Sussha et al. (2024) is closer to this approach, as students were tasked to co-write an argumentative essay with ChatGPT. However, the study explored students' reflections and experiences rather than studying expert assessment of the resulting essays. The second gap is that existing scholarly research is predominantly conducted with EFL and ESL university students (Bulqiyah et al., 2021; Jang et al., 2026; Lo et al., 2024; Mahapatra, 2024; Teng, 2024). Hence, a limited understanding of how ChatGPT influences underrepresented language contexts (Albadarin et al., 2024; F. Xiao et al., 2026). Scholars also reported that English prompts are more accurately responded to than those in other languages, particularly for complex reasoning (Lai et al., 2023; Rohera et al., 2025; Zhang et al., 2023). That said, the tool develops a better understanding of English structures, vocabulary, and context than in other, less popular languages. Little is known about the impact of ChatGPT on students' first-language academic writing, such as Indonesian, where language barriers are minimized, and students' cognitive and writing abilities can be more directly assessed.

Accordingly, the present study extends the line of inquiry by evaluating the actual essay quality written with ChatGPT assistance in the students' native language. The study employed a quantitative approach to understand whether ChatGPT's writing support is equally effective when used to write an essay in Indonesian. Students' essays were assessed by expert raters using the ESL Composition Profile (Jacobs et al., 1981), which is widely recognized for its robust and reliable analytic framework for evaluating university students' academic writing (Bacha, 2001; Ghanbari et al., 2012; Setyowati et al., 2020; Tiwari, 2023). As an analytic writing rubric, the Jacobs rubric evaluates five distinct dimensions of writing (content, organization, vocabulary, language use, mechanics), enabling a comprehensive assessment of both overall writing quality and individual components. The study combines between-group and within-group comparisons to examine whether access to ChatGPT was associated with higher essay scores under

assisted-writing conditions and to identify which writing dimensions benefit most from AI assistance. In addition, the study examines whether ChatGPT's effectiveness varies by students' initial writing performance to understand which learners benefit most from its use.

Based on these objectives, the study addressed the following research questions:

- RQ1. To what extent does ChatGPT assistance improve the quality of Indonesian argumentative essays?
- RQ2. Which dimensions of argumentative writing (content, organization, vocabulary, language use, mechanics) benefit most from ChatGPT assistance?
- RQ3. Does ChatGPT's effectiveness differ by students' initial writing performance?

Combining between-group and within-group comparisons, this study provides complementary evidence regarding the effect of ChatGPT on argumentative writing. Moving beyond average treatment effects by examining whether students with different initial levels of writing proficiency benefit differently from AI assistance, this study also contributes to the literature about ChatGPT's effectiveness.

METHODS

To examine whether ChatGPT significantly affected the writing performance of Indonesian university students when writing in their native language, this study employed a quantitative approach. The analysis first presented a comparison between two groups of students who wrote an essay on a similar topic under two conditions: without and with ChatGPT assistance. All students within the ChatGPT group used the free version of ChatGPT, which provided access to GPT-5. Later, the analysis presented a within-group comparison in which the group that wrote the essay without ChatGPT's assistance was tasked to write the same essay again with ChatGPT's assistance. Presenting this supplementary analysis allows observation towards changes in writing performance that may occur within the same students and examination towards which proficiency level of students is the ChatGPT's assistance more significant. While this

repeated-measures phase enabled the investigation of individual changes in writing performance before and after AI-assisted writing, maintaining the same topic, assessment rubric, time allocation, and testing procedures, the second time the students wrote the essay was intended to control for differences in topic difficulty and task characteristics. Hence, while the condition may cause a topic familiarity effect, writing performance can be examined under comparable conditions with minimal variability in task characteristics.

The research participants are 52 undergraduate students at an Indonesian private university majoring in IT who have prior experience writing academic essays in both English and Indonesian as part of their university coursework. Recruitment was conducted after securing access from the participating university. Prior to data collection, participants were informed of data confidentiality and security, and their consent was collected to ensure voluntary participation.

The participants were randomly assigned to two groups: one group of 25 students writing an argumentative essay without ChatGPT's assistance or any other GenAI tools, and the other of 27 students writing their essay with ChatGPT only and no other GenAI tools. The data for the between-group comparison were collected from the 25 students who completed the essay independently without ChatGPT assistance, and the 27 students who completed the same writing task with access to ChatGPT. Following the initial data collection, the 25 students in the non-ChatGPT group wrote the essay again using ChatGPT assistance.

All participants in both groups completed an argumentative essay in Indonesian responding to the prompt:

"Smartphone dan media sosial: apakah membantu atau menghambat produktivitas mahasiswa?"

In English, the prompt translates to 'Smartphone and social media: Do they help or hinder student productivity?' The topic was selected to resonate well with the student's

every day academic experiences. Smartphones and social media are no longer exclusive possessions; university students commonly use them for both educational and personal purposes. Hence, the essay topic reflects a realistic academic context, avoids specialized disciplinary knowledge, and reduces variability arising from unequal prior knowledge. Overall, the familiar topic supports students to formulate arguments, organize supporting evidence, and present a coherent position throughout the writing process, and allows assessment to focus more directly on argumentative writing ability.

Every student wrote their essay individually under similar supervised classroom conditions, with four supervisors present, within 60–90 minutes. The students assigned to the without-ChatGPT condition completed the essay independently, without access to ChatGPT or other GenAI tools, while students in the with-ChatGPT condition were allowed to access ChatGPT and receive any assistance from the application during their writing process.

Essay quality was evaluated using the ESL Composition Profile rubric (Jacobs et al., 1981). The rubric provides an overall writing score as well as scores for five dimensions: content, organization, vocabulary, language use, and mechanics. Two Indonesian lecturers with expertise in Indonesian language and writing instruction were assigned to assess every essay from both groups without being informed of which essay belonged to which group. Prior to assessment, all essays were cleared of any participant's personal data and given a code before being submitted to the raters. The average of the two raters' scores, both within each writing dimension and overall writing achievement, was used in all subsequent analyses to avoid rater bias.

Having two raters, this study examines ¹Inter-rater reliability using the Intraclass Correlation Coefficient (ICC) based on a two-way mixed-effects model with absolute agreement. The ICC values for each writing dimension and the overall writing score were calculated separately. The final scores were considered sufficiently reliable for subsequent statistical analyses, as the essays were independently

assessed by each rater using a common analytic rubric, and the two ratings were subsequently averaged.

The quantitative data analysis starts with descriptive statistics for each writing dimension and the overall writing score to inform the means and standard deviations.

Prior to conducting inferential analyses, the assumptions of normality and homogeneity of variance were examined using the Shapiro-Wilk and Levene's tests, respectively. Because homogeneity of variance was violated across several variables and the group sizes differed slightly between the groups with and without ChatGPT, Welch's independent-samples *t*-test was employed to compare participants' writing performance.

Changes in writing performance within the same students (the group without ChatGPT who later used ChatGPT for the second time) were examined using paired-samples *t*-tests.

To estimate the magnitude of the observed effects for both between-group and within-group comparisons, this study uses Hedges' *g*, given the sample's relatively small size. Statistical significance was determined at the .05 level. This study used IBM SPSS Statistics for all quantitative analyses.

RESULTS AND DISCUSSION

The consistency of the essay ratings between the two raters, as measured by ICC, is presented in Table 1.

Table 1 Inter-rater reliability of essay assessment.

Dimension	ICC	95% CI	Interpretation
Content	0.786	0.463-0.896	Good
Organization	0.65	0.381-0.793	Moderate
Vocabulary	0.466	0.155-0.622	Poor
Language Use	0.484	0.100-0.695	Poor
Mechanics	0.805	0.694-0.876	Good
Overall	0.601	0.150-0.790	Moderate

The ICC analysis presented in Table 1 shows variations of agreement between the two raters for the assessed writing dimensions and the

overall writing performance. The ICC analysis showed the highest level of agreement, hence good inter-rater reliability, for mechanics (ICC = 0.805, 95% CI = 0.694–0.876) and content (ICC = 0.786, 95% CI = 0.463–0.896). Moderate agreement was observed for organization (ICC = 0.650, 95% CI = 0.381–0.793) and the overall writing score (ICC = 0.601, 95% CI = 0.150–0.790), indicating acceptable consistency between the raters, but a comparatively lower agreement, thus greater variability in the raters' evaluations, was found for the remaining two dimensions; vocabulary (ICC = 0.466, 95% CI = 0.155–0.622) and language use (ICC = 0.484, 95% CI = 0.100–0.695).

Despite variability in the degree of agreement between raters, a common analytic rubric was adopted to assess the essay, and the average ratings from both raters were used for subsequent statistical analyses to provide a more stable estimate of students' writing performance while reducing the influence of individual rater variation. Hence, the ICC results generally indicate acceptable to good inter-rater reliability for the essay assessment, suggesting that the use of averaged scores in subsequent statistical analyses was acceptable. Having established the reliability of the essay ratings, the effectiveness of ChatGPT assistance on students' argumentative writing performance was examined through both between-group and within-group comparisons.

The first analysis examined the effect of ChatGPT assistance on students' argumentative writing performance, using a between-groups design. The essays from the first group ($n = 25$), who completed their essays without ChatGPT assistance, were compared with those from the second group ($n = 27$), who used ChatGPT's assistance. Given unequal group sizes and Levene's test indicating unequal variances across several writing dimensions, the between-group comparisons were conducted using Welch's independent-samples *t*-test, while Hedges' *g* was calculated to complement the statistical significance findings. Tables 2 and 3 present the descriptive statistics and inferential results of this analysis.

Table 2. Mean, Standard Deviation (SD), and Mean Difference for between-group comparison

Dimension	Without ChatGPT M (SD)	With ChatGPT M (SD)	Mean Difference
Content	23.78 (3.02)	25.65 (2.09)	1.87
Organization	16.08 (1.69)	17.15 (0.86)	1.07
Vocabulary	17.04 (1.15)	17.33 (1.00)	0.29
Language Use	19.96 (1.30)	20.63 (1.00)	0.67
Mechanics	3.80 (0.38)	3.89 (0.29)	0.09
Overall Score	80.50 (6.63)	83.98 (4.50)	3.48

Table 3. Independent-samples Welch's *t*-test and Hedges' *g* results for between-group comparison

Dimension	<i>p</i>	Hedges' <i>g</i>
Content	0.014	0.714
Organization	0.007	0.796
Vocabulary	0.334	0.265
Language Use	0.044	0.614
Mechanics	0.352	0.263
Overall Score	0.034	0.609

Tables 2 and 3 indicate that students who completed the essay with ChatGPT assistance scored higher than those who completed it independently, both overall and across all writing dimensions. Content ($p = .014$), organization ($p = .007$), language use ($p = .044$), and overall writing quality ($p = .034$) show statistically significant differences, unlike the other two dimensions: vocabulary ($p = .334$) and mechanics ($p = .352$). The magnitude of ChatGPT's influence, which varied across all writing dimensions, was further confirmed by the effect size analysis. The largest effect sizes were observed in organization ($g = 0.796$) and content ($g = 0.714$), indicating medium-to-large improvements in these dimensions with ChatGPT's intervention. A moderate effect was observed for language use ($g = 0.614$) and overall writing quality ($g = 0.609$), indicating moderate improvement in students' language use and final scores. Nevertheless, small effects were observed for vocabulary ($g = 0.265$) and mechanics ($g = 0.263$), indicating that ChatGPT's intervention had a negligible influence on these dimensions. Findings from the effect size analysis were consistent with the absence of statistically significant differences across all the assessed writing dimensions.

The analyses examining the effect of ChatGPT assistance on between-group comparisons indicate that students who used ChatGPT produce essays of significantly higher overall quality than those who wrote the essays independently, particularly in content development and organization. These findings suggest that lower-order linguistic features, such as vocabulary and mechanics, are less significantly affected by the adoption of GenAI in writing an argumentative essay.

Within-group analysis was conducted to complement the findings from the between-group comparison and provide sufficient insight into how effectively ChatGPT supports students in writing an argumentative essay. The within-group analysis was conducted among the 25 students who completed two argumentative essay-writing tasks on comparable topics: once independently and once with ChatGPT assistance. The analysis examines how students' writing performance changed after using ChatGPT as an assistant the second time they wrote the essay, across five writing dimensions and the overall essay score. Paired-samples *t*-tests were performed for this purpose as the same participants completed both writing tasks. Before conducting the paired-sample *t*-tests, normality of the difference scores was examined using the Shapiro-Wilk test, and the results show that the assumption was satisfied. Tables 4 and 5 provide the results from the descriptive statistics and paired-samples *t*-test.

Table 4. Mean, Standard Deviation (SD), and Mean Difference for within-group comparison

Dimension	Without ChatGPT M (SD)	With ChatGPT M (SD)	Mean Difference
Content	23.78 (3.02)	26.16 (0.93)	2.38
Organization	16.08 (1.69)	17.24 (0.84)	1.16
Vocabulary	17.04 (1.15)	17.72 (0.65)	0.68
Language Use	19.96 (1.30)	21.08 (0.70)	1.12
Mechanics	3.80 (0.38)	3.98 (0.10)	0.18
Overall Score	80.50 (6.63)	85.88 (3.00)	5.38

Table 5. Results of paired-samples *t*-tests for within-group comparison

Dimension	t	df	p	95% CI of Improvement	Effect Size
Content	3.68	24	0.001	[1.04, 3.72]	0.74
Organization	3.31	24	0.003	[0.44, 1.88]	0.66
Vocabulary	3.27	24	0.003	[0.25, 1.11]	0.65
Language Use	4.06	24	<.001	[0.55, 1.69]	0.81
Mechanics	2.22	24	0.036	[0.01, 0.35]	0.44
Overall Score	3.93	24	<.001	[2.56, 8.20]	0.79

Tables 4 and 5 show that ChatGPT has a positive impact on the quality of students' argumentative essays, as they achieve higher scores across all five writing dimensions and the overall writing score after using ChatGPT. The overall essay score shows a very significant improvement in students' scores after using ChatGPT, increasing from $M = 80.50$ ($SD = 6.63$) before ChatGPT assistance to $M = 85.88$ ($SD = 3.00$) after. This finding represents the statistically significant mean improvement of 5.38 points, $t(24) = 3.93$, $p < .001$. Among the writing dimensions, three show significant increases, with content leading the way with an average increase of 2.38 points, followed by organization (1.16 points) and language use (1.12 points). In contrast, vocabulary (0.68 points) and mechanics (0.18 points) show comparatively smaller improvements, despite the difference being statistically significant ($p = .036$).

Overall, the paired-samples analyses yielded moderate to large effects of ChatGPT assistance across the writing dimensions, with the largest effect observed for language use ($d_z = 0.81$). Overall score ($d_z = 0.79$) and content ($d_z = 0.74$) also show significant effects despite less than the effect on language use, but higher than organization ($d_z = 0.66$) and vocabulary ($d_z = 0.65$), which show moderate effects. The smallest effect was observed on mechanics ($d_z = 0.44$). Overall, the findings recommend that adopting ChatGPT can have a positive impact on students' argumentative writing across all dimensions, with greater impact on content and organization and less on vocabulary and mechanics.

To further understand the extent to which students of different proficiency levels benefit

from the ChatGPT assistance, this study used Pearson product-moment correlation analysis. The analysis was conducted on students' initial writing scores (scores obtained without ChatGPT) and their improvement scores (the difference between WithChatGPT and Without ChatGPT), as shown in Table 6.

Table 6. Pearson correlations between students' initial writing scores and improvement following ChatGPT assistance ($N = 25$).

Writing Dimension	Pearson's r	p
Content	-0.958	<.001
Organization	-0.881	<.001
Vocabulary	-0.831	<.001
Language Use	-0.864	<.001
Mechanics	-0.526	.007
Overall Score	-0.901	<.001

Comparing students' initial writing scores with the improvement following their use of ChatGPT's assistance, a statistically significant negative correlation across all writing dimensions was found (Table 6). The data suggests that ChatGPT's assistance is more influential for students with lower initial writing performance, as they experience greater improvement in their writing scores. The analysis shows the strongest negative correlation occurred for content ($r(23) = -.958$, $p < .001$) and the overall writing score ($r(23) = -.901$, $p < .001$), followed by organization ($r(23) = -.881$, $p < .001$), language use ($r(23) = -.864$, $p < .001$), and vocabulary ($r(23) = -.831$, $p < .001$). The very strong negative correlations across the five dimensions indicate that students whose performance was initially weaker on these dimensions achieved greater gains after using ChatGPT as an assistant during their writing process than their peers with stronger initial writing performance. Contrary to the five dimensions, mechanics showed a moderate negative correlation ($r(23) = -.526$, $p = .007$), indicating that students with weaker performance in the beginning still experience improvement in their mechanics assessment, but not as significant as the

improvement in the assessment gained for content, organization, vocabulary, language use, and overall score. The correlation analysis shows that students do not gain the same benefits from using ChatGPT as a writing assistant. Rather, the analysis recommends that ChatGPT's assistance is more effective for students with lower initial writing proficiency, as the data show that this group experiences greater improvements across nearly all dimensions of writing quality.

This study presents evidence showing that ChatGPT effectively improves the overall quality of argumentative essays, even when written in Indonesian. Contrary to previous studies (Lai et al., 2023; Rohera et al., 2025; Zhang et al., 2023), the present findings indicate that ChatGPT can respond accurately to languages other than English, despite its lack of representativeness in international linguistic scholarship. Hence, filling the gap for the limited understanding of how ChatGPT influences underrepresented language contexts (Albadarin et al., 2024; F. Xiao et al., 2026). Two complementary quantitative analyses from this study supported this interpretation. The between-group comparison showed that the withChatGPT group of students achieved significantly higher overall writing scores than their peers in the withoutChatGPT group, with a moderate effect size. Further, the within-group comparison showed that the same students can achieve higher scores when incorporating ChatGPT into their writing process.

Converging the findings across the two analytical approaches strengthens confidence that ChatGPT remains an effective tool for improving students' writing quality and aligns with Sussha et al.'s (2024) study. Hence, having prior integration of the tools for a certain period of time before seeing the benefits of ChatGPT's assistance in a writing task is not mandatory, as found in current literature (Alangari, 2025; Khampusaen, 2025; Mahapatra, 2024; Mai, 2026; Mekheimer, 2025; Mujeeb et al., 2026; Puteri et al., 2025).

Although the general findings of this study reinforce previous studies, regardless of national origin, recommending that ChatGPT

improve students' overall writing performance (Lo et al., 2024; Misbah Sultan et al., 2025; Nguyen et al., 2024; Werdiningsih et al., 2024), they also indicate that the improvement was not uniform across all aspects of writing. Some dimensions exhibited substantially greater gains than others, suggesting that ChatGPT did not influence every component of argumentative writing to the same extent as found in previous literature (Khampusaen, 2025; Mahapatra, 2024; Mai, 2026; Mekheimer, 2025; Mujeeb et al., 2026; Puteri et al., 2025; Q. Xiao, 2024). Understanding these differential effects remains essential for explaining how ChatGPT supports academic writing and for identifying the writing aspects most responsive to AI assistance.

Through both the between-group and within-group analyses, this study indicates that the greatest improvements were consistently observed in content and organization, followed by language use, whereas vocabulary and mechanics demonstrated comparatively smaller gains. These findings align with Mun's (2024) research in the Korean higher education context, but contrast with Alangari's (2025) study in Saudi Arabia, which found that the organization was less significantly affected, while vocabulary was more significantly improved. However, in terms of the mechanics least significantly affected by ChatGPT's intervention, this study aligns with Alangari's (2025) study. The convergence of these findings across two independent quantitative analyses suggests that ChatGPT was particularly effective in supporting higher-order aspects of argumentative writing, such as developing well-structured and coherent essays with clear content and flow of thoughts and ideas, while exerting a more limited influence on lower-order linguistic features, such as mechanics.

The significant improvement in content and organization represents one of the most important findings of the present study. The content quality of an argumentative writing reflects students' ability to generate ideas that are appropriate and relevant, construct arguments that are convincing, develop supporting evidence that is credible, and maintain the author's position clearly

throughout the essay. Such clear organization indicates how the two dimensions complement each other in the development of argumentative writing. Participants' significant score gains in the content and organization dimensions align with previous studies, particularly a systematic literature review that suggests ChatGPT can support students during the writing process by generating ideas and developing arguments (Lo et al., 2024; Misbah Sultan et al., 2025; Nguyen et al., 2024). Only this time, the present study shows that the effectiveness of prompts in Indonesian is equal, contributing to the existing literature in the Indonesian context, as ChatGPT assistance has already been found to be effective for Indonesian students' use of the tool for English essays (Puteri et al., 2025; Werdiningsih et al., 2024). Nevertheless, language use, despite indicating a smaller magnitude of improvement than the previous two dimensions, reflects the participants' grammatical accuracy and ability to express ideas convincingly and appropriately. The lower magnitude indicates that students' linguistic knowledge and ability to adapt ChatGPT's suggestions to their prompts remain important for maximizing the benefits of ChatGPT's assistance to their language quality.

In contrast, vocabulary and mechanics are the least affected dimensions, even with ChatGPT's assistance. As the between-group analysis indicates vocabulary shows a modest improvement, and the within-group analysis indicates a significant gain, this study recommends that when the same students revise their writing with ChatGPT, their scores are more likely to increase, indicating the significance of repeated interaction with AI during the writing and revision process. In contrast, the repeated interaction does not yield a significant effect, but it does show a small effect size for mechanics, even in the within-group comparison. This finding suggests that students may already perform relatively well in mechanics before using ChatGPT, leaving comparatively limited room for further improvement. This finding also suggests that ChatGPT's educational value in writing is more impactful, supporting the development

and communication of ideas rather than correcting surface-level writing features.

The correlation analyses further indicate that ChatGPT's effectiveness was not uniform across students of different proficiency levels. The students' scores on the first attempt, writing the essay without ChatGPT, are considered to reflect their initial proficiency in writing an argumentative essay. The analyses show that although all participants achieved higher scores after ChatGPT assistance (on the same essay in the same second attempt), the magnitude of improvement varied systematically by their initial writing proficiency. The significant negative correlations across all writing dimensions and the overall writing score confirm that students with lower initial scores generally experienced greater improvement than those with stronger initial writing performance. Aligning with Mekheimer's (2025), the evident patterns in the dimensions of content, organization, language use, and the overall writing score suggest that ChatGPT offered students who initially faced greater challenges in argumentative writing proportionally greater support.

These findings recognize ChatGPT's potential as a compensatory learning resource, helping reduce performance gaps among students with varying levels of writing proficiency. Hence, providing educators with alternatives to integrate ChatGPT to boost the achievement of low-achieving students in the classroom, as various studies have found that more students suffer from limited writing skills (Badi, 2015; Hartshorn et al., 2023; Mustafa et al., 2024; Nghi & Nhan, 2023; Ningrum et al., 2023). However, this finding should not be interpreted as evidence that ChatGPT is ineffective for higher-performing students. Rather, this finding may indicate that, specifically in this study, the relatively high-performing students had less room for measurable improvement within the writing assessment scale used. In contrast, students with weaker argumentative writing skills had greater scope for improvement within the writing assessment scale when assisted with ChatGPT. The stronger students initially show improvement in writing performance the second time they did the essay, but the magnitude of their gains

is comparatively smaller than that of their peers with lower skills.

The greatest improvement for the initially low-performing students was observed in content, the overall writing score, and organization. This finding aligns with the earlier analysis, which found that content and organization demonstrated the largest quantitative improvements when ChatGPT is adopted in the writing process. Overall, these findings suggest that ChatGPT's assistance is more valuable to students, especially less experienced writers who struggle to generate ideas, develop arguments, and organize information into coherent argumentative texts.

CONCLUSION

This study has presented evidence from both the between-group and within-group quantitative analyses indicating that ChatGPT exerts a meaningful and educationally important influence on the quality of argumentative essays written in Indonesian. Combining between-group and within-group analyses adds to the credibility of the data findings. The study consistently shows that with ChatGPT's assistance, students are likely to achieve higher overall writing scores than without it. Complementing the between-group analyses, the paired comparison further confirmed that the same students achieved significantly better performance after they used ChatGPT. Hence, these findings provide answers to the first research question, which inquires into the extent of ChatGPT's assistance in improving the quality of Indonesian argumentative essays.

The second research question, which inquires which dimensions of the students' argumentative writing are affected the most, is addressed by the findings from examining the effect sizes of their scores. The observed effect sizes, which ranged from moderate to large across the major writing dimensions, indicated that improvements were most evident in three dimensions, including content, organization, and language use, which can be categorized as higher-order writing dimensions. Meanwhile, relatively smaller gains were observed in vocabulary and mechanics, which are lower-

order linguistic features. These findings suggest that ChatGPT should not be perceived as a comprehensive solution for all aspects of writing quality but as a selective writing support tool that enhances idea generation, argument organization, and linguistic expression.

Finally, the Pearson product-moment correlation analysis comparing students' within-group writing scores before and after ChatGPT's assistance provides answers to the third research question, which inquires whether ChatGPT's effectiveness differs across students' initial writing performance. The analysis shows significant improvement across all writing dimensions for all students, but greater significance is evident among those with lower initial proficiency. These are the students who scored lower on their first writing attempt (without ChatGPT). Hence, ChatGPT assistance is likely more significant for students with lower initial writing skills, who often find academic writing challenging.

This study offers several implications regarding the integration of ChatGPT into higher education. Rather than positioning generative AI solely as a challenge to academic integrity, the findings suggest adopting GenAI from a pedagogical perspective that recognizes its potential as a scaffolding tool to enhance writing development when integrated responsibly into instruction. Hence, writing pedagogy should move beyond debates over whether students should use generative AI toward paying more attention to facilitating students with knowledge, skills, and AI literacy to evaluate AI-generated information critically, construct effective, ethical, and safe prompts, revise AI-generated suggestions appropriately, and maintain ownership of their academic work at all times. These are demanding skills in the current educational landscape as GenAI continues to evolve and becomes more deeply embedded in the academic environment.

This study still has some limitations worth recognizing. First, the transferability of the findings to other educational contexts, writing genres, or languages may be limited because this study was conducted within a single

Indonesian higher education context and focused exclusively on argumentative writing. Second, repeated administration of the same topic to within-group students may introduce a practice or topic familiarity effect. Therefore, precautions should be taken not to interpret the improvements observed in the second writing session as attributable solely to ChatGPT assistance. Finally, because the examination of students' writing performance was conducted once in a short period of AI-assisted writing without prior treatment, students' independent writing development is not observable.

Future studies may extend this work by using different research approaches to incorporate richer sets of data sources for analysis, including prompt histories to capture more directly how students interact with GenAI throughout the writing process. A future study may also adopt a longitudinal design to examine the consistency of improvements in students' writing proficiency associated with ChatGPT assistance over time and whether students' independent writing proficiency improves through the process when they finally write an essay without ChatGPT assistance. In addition, investigating ChatGPT's impact across different academic disciplines, writing genres, and educational contexts may enrich the present study's findings and help determine whether ChatGPT's influence is more significant in certain disciplines or academic backgrounds and less so in others.

Finally, this study suggests that ChatGPT's effectiveness should not be understood simply as a tool that merely improves writing quality, but as an educational tool whose educational values emerge through critical interaction and engagement between students and the tool. When used critically, strategically, and responsibly, ChatGPT has the potential to support the development of not only stronger argumentative writing skills but also other skills.

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